Row-Level Structural Integrity Diagnostics for Synthetic Tabular Data

Ahmed Fouad Lagha[1*], Zakarya Farou[1] and Imre Lendák[1]

[1*]Department of Data Science and Engineering, Eötvös Loránd University, Pázmány Péter sétány 1/C, Budapest, H-1117, Hungary.

*Corresponding author(s). E-mail(s): ahmed.lagha@inf.elte.hu; Contributing authors: zakaryafarou@inf.elte.hu; lendak@inf.elte.hu;

Abstract

Evaluating synthetic tabular data requires auditing individual record integrity in addition to global distribution alignment. Standard statistical metrics (KS complement, Total Variation Distance, Correlation Similarity) assess column distributions in aggregate and are fundamentally blind to localized, row-level conditional dependency violations. We present the Hybrid Integrity Framework (HIF), a row-level structural diagnostic that combines a Logical Sentinel Ensemble (LSE) for categorical conditional dependencies with Neighbor-Invariant Continuity (NIC) for continuous feature consistency. We demonstrate an integrative contribution: (1) **Empirically Robust Structural Diagnostics**: across tested generators (Gaussian Copula, Vine Copula, CTGAN, TVAE) and five benchmark datasets (Adult, Credit Default, Census ACS, Online Purchases, Supermarket Sales), HIF identifies row-level dependency ruptures even when aggregate marginal metrics report near-perfect fidelity (KS up to $\approx \mathbf{98\%}$); and (2) **Targeted Remediation**: pruning dependency-violating records ($\boldsymbol{H < 0.5}$) improves downstream predictive utility when structural errors are present (e.g., $\mathbf{+10.4}$ points F1 on Census ACS CTGAN, paired $\boldsymbol{t}$-test $\boldsymbol{p = 0.002}$, 95% CI $[\mathbf{+0.049, +0.160}]$; $\mathbf{+12.8}$ points on Online Purchases CTGAN, $\boldsymbol{p = 0.037}$, CI $[\mathbf{+0.010, +0.245}]$; the latter does not survive Bonferroni correction across 16 configurations) while preserving predictive utility in settings where structural errors are weak or absent (Gaussian Copula, paired $\boldsymbol{t}$-test $\boldsymbol{p = 0.13}$). HIF provides practitioners with a practical row-level signal of joint structural integrity for high-stakes synthetic data deployment.

Keywords: Synthetic tabular data; Data integrity; Row-level diagnostics; Dependency auditing; Conditional dependencies; Generative models

1 Introduction

The generation of high-fidelity synthetic tabular data has emerged as a cornerstone of reproducible research in privacy-sensitive domains such as economics [1], finance, and healthcare [2, 3]. With the scaling of deep generative architectures, such as CTGAN [4] and the refinement of flexible statistical models, such as Vine Copulas [5], the community has achieved remarkable success in matching aggregate marginal distributions. However, a critical "Dependency Fidelity Gap" remains: while these models excel at replicating per-column distributions, they frequently produce records that are individually plausible under each column's marginal yet violate the conditional inter-feature dependencies of the real-world joint distribution [6]—violations invisible to any metric that evaluates columns in isolation.

Traditional evaluation pipelines rely on *statistical fidelity* metrics, such as the Kolmogorov-Smirnov test and Total Variation Distance, to assess quality [7, 8]. While useful for verifying marginal distributional alignment, these metrics are inherently *aggregation-blind*: they evaluate each column's distribution independently and cannot detect records that violate conditional inter-feature dependencies yet remain within the expected marginal range of each individual column. Recent attempts to bridge this gap, such as the SHAP-based semantic fidelity metric [9], provide explainability-aware insights but often lack the scalability and technical rigor required for the release of high-stakes synthetic data. Crucially, this pathology is not exclusive to the opacity of deep neural networks: as we demonstrate, even rigorous statistical models (such as Vine Copulas) confidently hallucinate these joint dependency violations despite strong aggregate marginal fits.

To address these challenges, we present the Hybrid Integrity Framework (HIF), a row-level diagnostic system for detecting logical inconsistencies in synthetic tabular data. HIF is an integrative design that composes established building blocks—conditional classifiers, quantile-calibrated confidence thresholds, spectral embeddings, and mined association rules—into a single non-compensatory audit; its contribution lies in the problem formulation (per-record joint-conditional plausibility with record-level attribution) and in the systematic validation of when such auditing adds utility. HIF reframes the problem of fidelity evaluation as a multidimensional consistency detection task. By moving beyond aggregate statistical snapshots, the HIF identifies logical violations at the record level, enabling the filtering and curation of high-integrity synthetic cohorts. In Section 2, we contextualize our approach within the emerging literature on semantic fidelity and inter-column logic before detailing our methodology (calibration and auditing algorithms in the Appendix).

Our framework comprises three complementary auditing layers:

1. **Logical Sentinel Ensemble (LSE):** An array of discriminative classifiers that identify high-dependency categorical feature hubs in the ground-truth data using cross-validated classification accuracy.
2. **Neighbor-Invariant Continuity (NIC):** A continuous feature auditing mechanism that verifies local semantic adjacency within SVD-embedded categorical contexts using robust percentile thresholding.

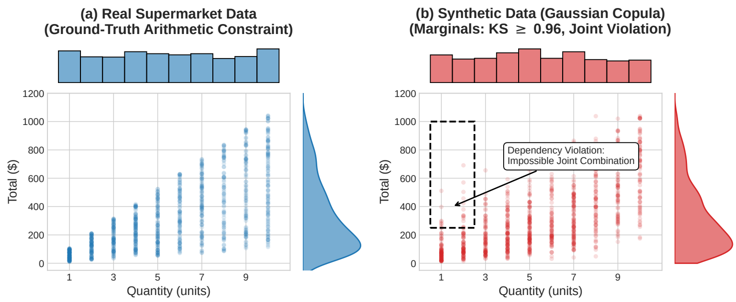

Fig. 1 An empirical demonstration of the Dependency Gap on the Supermarket Sales dataset. While generative models (e.g., Gaussian Copula) successfully replicate the 1D marginal distributions of *Quantity* and *Total* (side histograms; per-column KS ≈ 0.96 for these two marginals, versus the dataset-level KS of 0.903 ± 0.006 reported in Table 1), they fail to preserve the joint arithmetic constraint (Total $=$ Price $\times$ Quantity $\times$ $(1 + \text{Tax})$), hallucinating impossible records (dashed box) that aggregate marginal metrics do not capture.

3. **Automated Rule Extractor:** An association rule mining engine that extracts high-confidence deterministic implication rules ($A \implies B$) from ground-truth data to enforce hard structural constraints.

Summary of Contributions.

HIF contributes (i) an **Integrative Row-Level Structural Diagnostic**: by composing predictive-synergy hub selection, quantile-calibrated conditional classifiers, robust spectral-continuity auditing, and mined implication rules into one framework, it reframes synthetic data auditing from aggregate column matching to record-level conditional plausibility with non-compensatory multiplicative aggregation ($\prod_l(1-\mathcal{P}_l)$) to detect localized structural hallucinations ("impossible junctions") where every individual column value appears marginal-plausible yet their joint combination violates conditional logic; and (ii) **Targeted Remediation**: we characterize when integrity filtering improves downstream utility—on structural-error cohorts (e.g., CTGAN on Census ACS), pruning low-HIF records ($H < 0.5$) yields statistically significant gains (+10.4 points F1, paired t-test $p = 0.002$, 95% CI $[+0.049, +0.160]$; +12.8 points on Online Purchases with CTGAN, $p = 0.037$, CI $[+0.010, +0.245]$; this effect does not survive Bonferroni correction across 16 configurations), while on largely error-free cohorts (e.g., Gaussian Copula on Census ACS) filtering preserves utility without detectable degradation (paired t-test $p = 0.13$).

By providing a scalable, data-driven diagnostic, HIF enables the production of synthetic data certified for high-stakes research. The complete open-source implementation, experimental benchmarks, and dataset configurations are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

2 Related Work

Synthetic Tabular Data Generation.

The field has evolved from early copula-based methods to deep generative architectures. GAN-based models such as CTGAN and TVAE [4] set an early benchmark for capturing joint distributions by building on the Synthetic Data Vault framework [10], later improved for skewed distributions by CTAB-GAN [11]. Diffusion paradigms such as TabDDPM [12], flow-based gradient-boosted tree generators [13], and composable architectures such as CoDi [14] have pushed the frontier of manifold matching; permutation-invariant objectives [15] and federated frameworks [16] extend synthesis to column-order robustness and decentralized settings. However, both statistical and deep architectures suffer a fundamental objective mismatch: statistical models (such as Vine Copulas) excel at per-column marginal fitting but smooth over complex non-linear conditional constraints, while neural models (such as CTGAN) can learn non-linear categorical contexts but remain prone to mode collapse or localized joint violations. Our work focuses on a representative spectrum of generators—from classical *Gaussian Copulas* [17] and *Vine Copulas* [5] to neural architectures such as *CTGAN*—and shows that the Dependency Gap recurs across these diverse generative paradigms in our tested settings.

Evaluation Metrics and the Dependency Gap.

Traditional evaluation pipelines primarily rely on statistical fidelity (e.g., Kolmogorov-Smirnov tests, Correlation Distance) and predictive utility (Train-on-Synthetic, Test-on-Real; TSTR). Although effective for global assessment, these metrics evaluate aggregate column-level statistics and are therefore blind to row-level conditional dependency violations—by design, a corruption that samples each feature from its correct marginal but ignores inter-feature correlations is entirely invisible to such metrics. Long et al. (2025) [6] were among the first to formalize these "Dependency Gaps," proposing rule-based audits for inter-column relationships. Umesh et al. [18] studied the preservation of logical and functional dependencies from a generation-side perspective, constraining the generative model to respect mined dependencies at training time; this is complementary to—but distinct from—a post-hoc audit of an arbitrary generator's output, which is the setting we address, and dependency-aware representations have recently been shown to enhance tabular understanding [19]. Similarly, Yu et al. (2025) [9] introduced the SHAP Distance, an explainability-aware fidelity metric, but it yields a dataset-level attribution summary rather than per-record joint-conditional scores and scales exponentially with the feature count (see Appendix D.3).

Sample-Complexity Limits on Aggregate Auditing.

Distribution-level auditing additionally faces provable sample-complexity limits: identity and uniformity tests require sample sizes that scale with the domain support [20–22], a cost only partially mitigated even for product-structured distributions [23], and conditional independence testing is likewise sample-intensive in high dimensions [24]. These bounds constrain any aggregate audit of large categorical spaces and motivate scoring records individually rather than testing the synthetic cohort globally.

They are also computable at HIF's own operating configuration (Section 3): inverting the identity-testing requirement $m \gtrsim \sqrt{S}/\varepsilon^2$ [20, 22] at the per-hub, per-category sample sizes HIF actually conditions on (2,000 rows, $\leq$ 10 features, top-5 hubs) yields per-cell detectability floors of $\varepsilon \approx 0.27$–1.0 across our five benchmarks versus a cohort-level analogue of 0.15–0.19 (Appendix D.2). This formalizes the granularity below which an individual sentinel cell cannot certify deviations and motivates the confidence-floor safety minimum and the multi-hub geometric aggregation of Section 3, which dampen the long-tail cells that drive the worst-case floor.

HIF automates dependency discovery through predictive synergy scoring: unlike rule-mined audits, which are binary and require manual specification, HIF provides a soft probabilistic integrity score using non-compensatory multiplicative aggregation that prevents high-fidelity feature dimensions from masking localized dependency violations.

3 Mathematical Foundation

Terminology.

Throughout, four related but distinct terms are used. *Fidelity* denotes aggregate distributional alignment—how closely a synthetic cohort's marginal or pairwise statistics match the real data—as measured by metrics such as KS, TVD, and JCD. *Integrity* is the row-level property this framework audits: a record has high integrity if its joint configuration is consistent with the conditional dependencies learned from the real data, formalized as the HIF score $H(\mathbf{x})$. *Plausibility* is weaker and per-column: a record is plausible if each of its values falls within the expected marginal range of its feature, which a dependency-violating record can still satisfy. *Validity* refers to whether a record is actually possible under the true data-generating process; because HIF certifies consistency with a *learned* conditional manifold, a low-integrity record is best read as "structurally inconsistent with the training distribution" rather than universally invalid (Section 4).

The Hybrid Integrity Framework (HIF) formalizes synthetic data auditing as a probabilistic conditional scoring task. Let $\mathcal{D}_{real}$ denote the original training dataset. We define the **ground-truth manifold** $\mathcal{M}_{real}$ as the support of the joint distribution $P_{real}(\mathbf{x})$—the localized subspace of valid feature configurations that occur in reality [25]. A synthetic record $\mathbf{x} = [\mathbf{x}_{cat}, \mathbf{x}_{cont}]$ (with $\mathbf{x}_{cat}$ and $\mathbf{x}_{cont}$ the sub-vectors of categorical and continuous features) is a *dependency-violating record* if it is plausible under each marginal distribution yet falls outside the ground-truth manifold (i.e., it is highly improbable under the joint conditional distribution $P_{\mathcal{D}_{real}}(\mathbf{x}_h \mid \mathbf{x}_{\setminus h})$ learned from the data). This distinction is central: aggregate fidelity metrics (KS, TVD, JCD) assess whether records fall within the expected marginal range of each column in isolation, whereas HIF additionally evaluates whether each record's joint configuration structurally aligns with the ground-truth manifold. The complete calibration process for establishing these data-driven manifold boundaries is detailed in Algorithm 1.

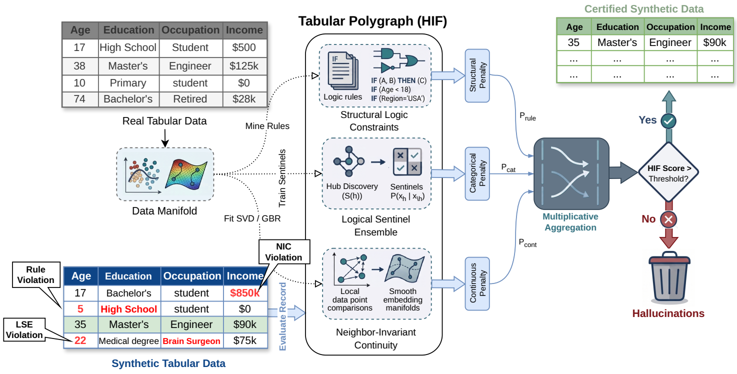

Fig. 2 Architecture of the Hybrid Integrity Framework (HIF). Synthetic tabular data is verified by the Logical Sentinel Ensemble (LSE) and Neighbor-Invariant Continuity (NIC) modules, combined via multiplicative aggregation to filter semantically inconsistent records.

3.1 Logical Sentinel Ensemble (LSE)

To audit categorical consistency, we first identify the set of Dependency Hubs $\mathcal{H}$ that govern the structural dependencies of the dataset. For a given row $\mathbf{x}$, let $x_h \in \mathcal{X}_h$ denote the categorical value of a target feature h, and let $\mathbf{x}_{\setminus h}$ denote the values of all other features (i.e., $\mathbf{x}$ with feature h removed).

We train an independent Random Forest classifier [26]—leveraging decision tree recursive partitioning to capture non-linear feature interactions [27]—which we call a *Sentinel* and denote as f_h, to predict x_h given $\mathbf{x}_{\setminus h}$. The sentinel outputs a conditional probability distribution $P_{f_h}(x_h = c \mid \mathbf{x}_{\setminus h})$ for all possible categories $c \in \mathcal{X}_h$. We define the importance of a feature h using the Feature Dependency Score $S(h)$, the cross-validated classification accuracy of its sentinel:

$$S(h) = \mathbb{E}_{\mathbf{x}\sim\mathcal{D}_{real}}\left[\mathbb{I}\left(\arg\max_{c\in\mathcal{X}_h} P_{f_h}(x_h = c \mid \mathbf{x}_{\setminus h}) = x_h\right)\right] \tag{1}$$

where $\mathbb{I}$ is the indicator function. By maximizing $S(h)$, we select hubs that are most strongly constrained by conditional interactions with other attributes. While computationally efficient, this reliance on single-variable classification accuracy primarily captures univariate dependencies and may under-represent higher-order parity relationships; future work will investigate Total Correlation and multi-way interaction kernels to broaden hub selection.

Consider, for illustration, a demographic dataset where x_h is *Education Level* and $\mathbf{x}_{\setminus h}$ contains *Age* (e.g., 14). Because a 14-year-old cannot hold a PhD in the real world, the sentinel f_h learns from the ground-truth data to assign $P_{f_h}(\text{PhD} \mid \text{Age} = 14) \approx 0$. If a generative model hallucinates a 14-year-old with a PhD, the sentinel will flag the record because the assigned probability for the hallucinated category is extremely low.

Confidence Floor Calibration.

Drawing inspiration from distribution-free quantile calibration in inductive conformal prediction [28–30], we establish a data-driven Confidence Floor δ_h for each hub based on empirical calibration error bounds, defined as the empirical 5th-percentile ($Q_{0.05}$) of the *out-of-bag* probability mass assigned to the correct category by the sentinel in the ground-truth data, with a safety minimum of 0.01:

$$\delta_h = \max\left(Q_{0.05}\left(\{P_{f_h}(x_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}\right), 0.01\right) \tag{2}$$

Using out-of-bag predictions (i.e., predictions of the trees that did not train on each row) yields out-of-sample floors without requiring a held-out split, preventing over-optimistic calibration on rows the sentinels memorized during training. This 5th-percentile floor acts as a conservative safety margin: if the real data occasionally contains 18-year-olds with college degrees, the sentinel's probability might be low but non-zero (e.g., 0.05), and the floor δ_h ensures we do not penalize rare but valid records that fall within the natural density of the ground-truth manifold.

The per-hub penalty $\mathcal{P}_h(\mathbf{x})$ uses a soft-threshold activation to suppress noise-level deviations, harshly penalizing only those records whose probabilities fall far below the confidence floor (such as the impossible 14-year-old PhD):

$$\mathcal{P}_h(\mathbf{x}) = \text{clip}\left(\frac{\delta_h - P_{f_h}(x_h \mid \mathbf{x}_{\setminus h})}{\delta_h}, 0, 1\right) \tag{3}$$

The categorical penalty aggregates multiplicatively across hubs, ensuring that ruptures in multiple manifold dimensions are compounded.

$$\mathcal{P}_{cat}(\mathbf{x}) = 1 - \prod_{h \in \mathcal{H}} (1 - \mathcal{P}_h(\mathbf{x})) \tag{4}$$

The complete manifold calibration procedure—hub selection, confidence-floor calibration, NIC regressor training, and rule mining—is summarized in Algorithm 1 (Appendix B).

3.2 Neighbor-Invariant Continuity (NIC)

For continuous features $\mathbf{y}$, we evaluate integrity via Semantic Adjacency. We map the categorical context to a continuous latent space $\mathcal{Z}$ using a standardized spectral projection $\phi : \mathcal{X}_{cat} \to \mathcal{Z}$—drawing on classical Multiple Correspondence Analysis (MCA) and Singular Value Decomposition (SVD) on binary indicator matrices [31, 32]. Standardizing the indicator matrices by their standard deviation prior to SVD mathematically replicates the inverse-frequency weighting of MCA, ensuring that rare but highly informative categorical constraints dominate the continuous manifold projection. We then model the conditional expectation of each continuous feature y as a non-linear function of this manifold using a Gradient Boosting Regressor r_y [33, 34]:

$$\hat{y}(\mathbf{x}) = \mathbb{E}[y \mid \phi(\mathbf{x}_{cat})] \approx r_y(\phi(\mathbf{x}_{cat})) \tag{5}$$

For any given record $\mathbf{x}$, let $e_y(\mathbf{x}) = |y(\mathbf{x}) - \hat{y}(\mathbf{x})|$ denote the absolute regression residual, and define the ground-truth residual set as $\mathcal{E}_y = \{e_y(\mathbf{x}) \mid \mathbf{x} \in \mathcal{D}_{real}\}$. For example, a synthetic 45-year-old Senior Executive generated with an Income of \$12 yields an enormous residual against the expected executive income ($\hat{y} \approx$ \$120,000), far exceeding the natural variance of executives in the real dataset and signaling a continuous dependency gap. To handle extreme feature skewness (common in census domains), we define a robust hybrid threshold τ_y from $\mathcal{E}_y$ using the Median Absolute Deviation (MAD) [35], a robust estimate of local manifold spread that is less sensitive to outliers than the standard deviation:

$$\tau_y = \max\left(Q_{0.95}(\mathcal{E}_y), \text{Median}(\mathcal{E}_y) + 2 \cdot \text{MAD}(\mathcal{E}_y)\right) \tag{6}$$

This threshold keeps the NIC layer stable even in high-skew distributions, such as macroeconomic wage data.

The continuous penalty $\mathcal{P}_{cont}(\mathbf{x})$ uses a dynamic scaling factor $\gamma_y = \max(Q_{0.98}(\mathcal{E}_y), 2\tau_y, 3\text{MAD}(\mathcal{E}_y), 0.1)$ that bounds the penalty relative to the feature's natural noise floor, so small deviations beyond τ_y yield soft penalties while massive violations (like the \$12 executive) yield a maximum penalty of 1.0:

$$\mathcal{P}_{cont}(\mathbf{x}) = \max_y \left[\text{clip}\left(\frac{e_y(\mathbf{x}) - \tau_y}{\gamma_y}, 0, 1\right)\right] \tag{7}$$

As a defensive guard, NIC detects *collapsed* continuous features whose conditional predictions have near-zero variance (i.e., the feature is effectively constant given the categorical context); for such degenerate features, all records receive a uniform base penalty since the expected residual distribution provides no reliable baseline. This guard did not activate on any benchmark dataset in our experiments.

3.3 Multiplicative Aggregation

In contrast to additive fidelity metrics that allow a model to "compensate" for logical failures with high statistical overlap, HIF formalizes integrity as an Algebraic Intersection: the hybrid integrity score $H(\mathbf{x})$ is defined as the geometric mean of the component validities:

$$H(\mathbf{x}) = \left(\prod_{l \in \{cat, cont, rule\}} (1 - \mathcal{P}_l(\mathbf{x}))\right)^{1/L} \tag{8}$$

where l iterates over the active auditing layers. The geometric mean preserves the non-compensatory property of a Logical AND gate: a significant failure in any single manifold dimension ($\mathcal{P}_l \to 1$) drives the entire record integrity toward zero, regardless of how well the other components align (see Table 6, Section 4.5). For a formal analysis of this aggregation logic under explicit assumptions—intersection soundness (Theorem 1), asymptotic convergence (Theorem 2), and the Type I false-rejection bound under quantile calibration (Theorem 3)—we direct the reader to **Appendix A**.

While HIF prioritizes soft probabilistic validities via LSE and NIC, we incorporate symbolic implication rules as an optional *Structural Logic Constraint* layer ($l = rule$) to enforce known deterministic or physical laws (e.g., Age ≥ 18 for voters), bridging differentiable manifolds and rigid symbolic constraints.

The Integrity Frontier.

The dataset-level HIF score is the expected record-level integrity. To provide a discrete diagnostic, a record is classified as a *Semantic Violation* if $H(\mathbf{x}) < 0.5$, defining the framework's Integrity Frontier.

$$\text{HIF Score} = \frac{1}{N}\sum_{i=1}^{N} H(\mathbf{x}_i) \tag{9}$$

This score provides a bounded diagnostic $\in [0, 1]$, where 1.0 indicates a logically certified synthetic cohort. The complete end-to-end auditing procedure for scoring a synthetic dataset is detailed in Algorithm 2 (Appendix B).

Because HIF penalizes records that violate *learned conditional dependencies* rather than marginal rarity—the confidence floor δ_h is calibrated on the ground-truth manifold precisely to avoid penalizing rare-but-valid tail records—filtering operates on structural consistency rather than demographic support; a systematic audit of representation drift across sensitive attributes under integrity filtering is left to future work. A primary critique of multiplicative evaluation is that absolute scores often appear lower than the near-unity scores (≥ 0.95) typical of aggregate 1D distributional metrics in standard evaluation benchmarks [2, 6, 10]. This is a deliberate design property: a record with a severe violation of any single dependency should not receive a high integrity score, regardless of how well its remaining features conform to marginal expectations. Assuming conditional independence of error dimensions given the feature representations, the product of validities $\prod(1 - \mathcal{P}_l)$ serves as a pseudo-likelihood proxy for the joint structural conformity of the record, and under empirical quantile calibration ($\delta_h = Q_\alpha$, $\gamma_y = Q_{1-\beta}$) the false-rejection probability for valid tail records is upper-bounded by the Union Bound under stated assumptions (Theorem 3, Appendix A).

4 Results

We evaluated the Hybrid Integrity Framework (HIF) on five benchmark datasets with diverse structural complexity: U.S. Census ACS [36, 37], Adult Income, Credit Card Default, Online Purchases, and Supermarket Sales. Experiments used $N = 3$–10 random seeds depending on the evaluation; we report mean $\pm$ standard deviation (SD) unless otherwise noted.

To calibrate the framework's response, we executed a controlled semantic violation protocol: we stochastically permuted (shuffled) the values of specific features across different rows within highly constrained categorical-continuous dependencies for a controlled fraction of records $\eta \in [0, 0.6]$ (Census ACS, $N = 3$ seeds). Shuffling existing values across rows rather than injecting external noise preserves each column's marginal distribution (satisfying standard MM and KS metrics) while breaking the underlying joint conditional structure at the row level. Under this protocol, the HIF score exhibited strong monotonicity in our runs (Spearman $\rho = -0.98$ across seeds): the mean HIF decreased from 0.941 ± 0.003 at $\eta = 0$ to 0.712 ± 0.002 at $\eta = 0.6$, whereas the marginal metrics—Moment Matching (MM) and Kolmogorov-Smirnov (KS)—remained effectively static ($\rho \approx 0$), since the shuffled values are drawn from the intact marginals. Notably, the Joint Correlation Distance (JCD) also decayed sharply ($88.8 \rightarrow 65.0$, $\rho = -0.98$): shuffling a feature destroys its pairwise correlations, so JCD is a sensitive aggregate drift detector—but it yields only a dataset-level summary and cannot attribute degradation to specific records. This motivates HIF's row-level output and the low-rate, marginals-preserving conditional-swap protocol used in Section 4.4.

Evaluation Protocol and Threats to Validity.

We distinguish ranking quality (ROC-AUC, PR-AUC) from thresholded classification quality (F1), treating the former as threshold-independent diagnostics of error ranking and the latter as an operating-point metric. For baseline comparability, IF and LOF use a `contamination` parameter set to the known corruption rate, and the learned-density baseline (GMM) uses the corresponding quantile of real-data scores, in synthetic stress tests; this favors those baselines for F1 and is reported explicitly to avoid hidden calibration advantages (we likewise report HIF and GMM at a matched operating point in Section 4.3). The controlled corruption protocol above is used strictly for metric response calibration, not as the primary benchmark for diagnostic utility: our main empirical validation assesses HIF across three independent regimes—(1) a cross-architecture maturity audit of generative models (Section 4.1), (2) downstream predictive utility on un-manipulated synthetic cohorts (Section 4.2), and (3) held-out error families that HIF was not specifically engineered to detect (plus the targeted dependency-violation family as a positive control; Section 4.3). Sample sizes differ by experiment: the significance, held-out, and cross-architecture audits use $N = 10$ seeds with paired tests and exact p-values where applicable, whereas the calibration-domain experiments use $N = 3$ seeds and are reported as descriptive mean $\pm$ SD without inferential claims; the calibration-domain and statistical-significance analyses are confined to Census ACS, so their inferential conclusions should not be extrapolated to other domains without further evaluation. Residual threats include dataset selection bias and the possibility that rare-but-valid tail patterns are partially coupled with high-penalty regions. Two caveats bound the construct validity of the "violation" label. First, HIF certifies inconsistency with the *learned* conditional manifold: a flagged record need not be impossible, only sufficiently improbable under the conditional dependencies estimated from the real data, so the frontier should be read as "structurally inconsistent with the training distribution" rather than "universally invalid." Second, where invalidity is externally verifiable—the Supermarket Sales arithmetic identity (Total = Price $\times$ Quantity $\times$ (1 + Tax))—the flagged cohort was confirmed to violate the constraint, providing an independent anchor for the penalty; such verifiable constraints are the strongest evidence for the hallucination interpretation and should be prioritized when available.

4.1 Cross-Architecture Maturity Audit and Cohort Certification

To evaluate the diagnostic utility of HIF across generative paradigms, we conducted a cross-architecture audit across four benchmark datasets from diverse domains (Table 1): Supermarket Sales (retail arithmetic constraints), Online Purchases (e-commerce transaction constraints), Credit Card Default (financial classification), and Adult Census Income (demographic benchmark). The results reveal a key insight—an *Integrity-Fidelity Decoupling*: a generator can match aggregate marginal statistics (KS) while simultaneously violating the joint conditional structure of the data, so that high marginal fidelity does not imply row-level integrity, and vice versa:

1. Integrity-Fidelity Decoupling on Constrained Domains.

Across all four datasets, Vine Copulas achieve high marginal statistical fit (KS $\geq$ 0.969). However, on datasets with cross-feature relational constraints (Supermarket Sales and Online Purchases), Vine Copulas exhibit severe joint dependency failures, with violation rates of **86.2% to 99.9%** (HIF $\leq$ 0.271); CTGAN and TVAE show similar deficits on transaction constraints (HIF = 0.02–0.28, violation rates 84.9%–100%). Marginal metrics (KS) report these models as high-fidelity, underscoring that aggregate fidelity metrics do not capture joint dependency ruptures. A qualitative audit on Supermarket Sales (Vine Copula) confirmed the mechanism: generators match each column's marginal distribution yet violate the arithmetic identity subtotal = unit_price × quantity × (1 + tax), with HIF flagging essentially the entire cohort (violation rate 99.9% across $N = 10$ seeds; component rates 100.0% LSE, 63.2% rules, 0.0% NIC)—while KS and joint-correlation metrics reported high fidelity. This identity is externally verifiable, independent of the auditor's learned models, and anchors the construct validity of the HIF penalty in a way that learned conditional improbability alone cannot (see the threats-to-validity discussion in Section 4).

These violations are independently confirmable beyond the auditor's own learned models. Using arithmetic identities that hold to $\approx 10^{-13}$ relative error in the real data—Total = Unit Price × Quantity × 1.05 (equivalently Total = cogs + gross income) on Supermarket Sales, and item_total = item_subtotal + item_tax together with item_subtotal = purchase_price × quantity on Online Purchases—we verified every real generator cohort in this audit ($N = 10$ seeds). Records flagged by HIF ($H < 0.5$) are independently confirmed as identity-violating in $\geq$ 98.9% of cases across all eight dataset–generator configurations (1.000 in six of eight; Table 2). The constraints are violated wholesale (base violation rate 0.87–1.00), so on these fully constrained domains the identity check confirms the Dependency Gap rather than discriminating among records: the cleanest discriminator is Gaussian Copula on Online Purchases, where unflagged records violate at 0.720 versus 0.989 among flagged, and the HIF penalty correlates with violation severity (Spearman $\rho = 0.47 \pm 0.01$; Table 2). Conversely, where HIF flags essentially the entire cohort (Vine and CTGAN on Supermarket Sales; Vine, CTGAN, and TVAE on Online Purchases), its penalty is only weakly monotone in arithmetic severity ($\rho \leq 0.18$), because the cohort's failures are dominated by categorical-structure violations rather than continuous residuals—precisely the regime in which the learned-density baseline of Section 4.3

is the sharper detector of the arithmetic constraint (GMM ROC-AUC 0.955 vs. 0.797 on Online Purchases). These numbers quantify the earlier qualitative audit: the flagged cohort is externally invalid at near-universal rates, while HIF's *ranking* within arithmetic-constraint domains is limited—a boundary that motivates the complementary learned-density baseline reported in Section 4.3.

2. Evaluation on Unconstrained Manifolds.

On unconstrained demographic and financial benchmarks (Adult Income, Credit Default), raw generators preserve conditional structures better, with HIF scores ranging from 0.33 to 0.98 (violation rates 0.5%–67.6%), providing a continuous gradient of structural integrity for comparing generator maturity. Notably, TVAE achieves the highest HIF on both Credit Default (0.827) and Adult Income (0.981), demonstrating that VAE-based architectures can excel at preserving joint dependencies on unconstrained domains.

3. Quantitative Decoupling Evidence.

Pearson correlation analysis across all 16 dataset–generator configurations of the four audit benchmarks (Census ACS is held out as the calibration benchmark; Table 1) characterizes the decoupling precisely: HIF shows no linear relationship with marginal fidelity ($\rho(\text{HIF}, \text{KS}) = 0.038$) and only a weak relationship with support coverage ($\rho(\text{HIF}, \alpha) = 0.325$), but a strong positive correlation with joint correlation fidelity ($\rho(\text{HIF}, \text{JCD}) = +0.82$). HIF thus aligns with joint-dependency structure while being decoupled from per-column marginal metrics: high marginal fidelity (KS) does not imply high integrity, exactly the pattern observed for Vine Copulas on constrained domains.

4.2 Alignment with Downstream Utility

A critical requirement for any synthetic data metric is its ability to act as a proxy for downstream predictive utility, a validation paradigm now standard in privacy-sensitive domains [38]. We measured downstream TSTR F1 under integrity filtering across representative generative architectures (Table 3), using a Random Forest classifier [26] (via *Scikit-learn* [39]); the *Rule-only Baseline* prunes records violating any mined implication rules, whereas the *HIF Filtered* variant uses the semantic frontier ($H < 0.5$).

The paired tests on Census ACS ($N = 10$ seeds, binary median-split `household_income` target) quantify the asymmetry. For the Gaussian Copula, filtering yields $p = 0.133$ (95% CI $[-0.008, +0.001]$): no detectable utility degradation, as expected for a cohort largely free of conditional dependency violations (violation rate $< 1\%$). For CTGAN, downstream utility gains are statistically significant (paired t-test $p = 0.002$; Wilcoxon $p = 0.002$; 95% CI $[+0.049, +0.160]$; paired Cohen's $d \approx 1.34$), with F1 improving from 0.427 ± 0.090 to 0.532 ± 0.139 ($+0.104$), supporting the claim that HIF is particularly useful for architectures that produce joint dependency violations. The *Rule-only Baseline* isolates the symbolic layer: pruning the records that violate mined implication rules ($\approx 3\%$ of the cohort; retention 97.0%) leaves utility unchanged (Census ACS CTGAN: F1 0.426 vs. 0.427 unfiltered; Gaussian Copula:

Table 1 Cross-Architecture Audit: Statistical Fidelity (KS), Support Overlap (α-Precision, β-Recall), and Joint Dependency Integrity (HIF) across benchmark datasets ($N = 10$ seeds, Mean $\pm$ SD). High marginal fidelity (KS > 0.96) or high support coverage (α-Prec > 0.79) does not imply joint conditional integrity. On datasets with cross-feature constraints, models exhibit up to 100% dependency violation rates.

Generator	KS (↑)	α-Prec (↑)	β-Rec (↑)	HIF (↑)	Viol. Rate (↓)
Supermarket Sales					
Gaussian Copula	0.903 ± 0.006	0.892 ± 0.012	0.918 ± 0.013	**0.963 ± 0.006**	**2.6 ± 0.6%**
Vine Copula	**0.975 ± 0.003**	0.792 ± 0.011	0.862 ± 0.016	0.025 ± 0.002	99.9 ± 0.1%
CTGAN	0.771 ± 0.029	0.644 ± 0.071	0.861 ± 0.027	0.019 ± 0.010	100.0 ± 0.0%
TVAE	0.881 ± 0.011	0.827 ± 0.038	0.953 ± 0.015	0.021 ± 0.001	100.0 ± 0.0%
Online Purchases					
Gaussian Copula	0.753 ± 0.010	0.489 ± 0.013	0.819 ± 0.023	**0.626 ± 0.009**	**40.3 ± 1.3%**
Vine Copula	**0.969 ± 0.004**	0.831 ± 0.019	0.942 ± 0.013	0.271 ± 0.008	86.2 ± 1.3%
CTGAN	0.677 ± 0.068	0.524 ± 0.061	0.882 ± 0.047	0.278 ± 0.047	84.9 ± 5.7%
TVAE	0.854 ± 0.019	0.754 ± 0.049	0.818 ± 0.035	0.119 ± 0.008	99.0 ± 0.5%
Credit Default					
Gaussian Copula	0.747 ± 0.002	0.729 ± 0.009	0.951 ± 0.010	0.612 ± 0.005	30.5 ± 0.8%
Vine Copula	**0.981 ± 0.001**	0.801 ± 0.011	0.813 ± 0.014	0.401 ± 0.007	55.7 ± 1.1%
CTGAN	0.717 ± 0.059	0.627 ± 0.066	0.859 ± 0.018	0.333 ± 0.038	67.6 ± 4.8%
TVAE	0.878 ± 0.005	0.961 ± 0.009	0.935 ± 0.007	**0.827 ± 0.009**	**13.5 ± 1.1%**
Adult Income					
Gaussian Copula	0.848 ± 0.006	0.939 ± 0.006	0.838 ± 0.016	0.830 ± 0.007	10.5 ± 0.8%
Vine Copula	**0.981 ± 0.005**	**0.964 ± 0.008**	0.799 ± 0.021	0.764 ± 0.004	15.9 ± 0.9%
CTGAN	0.682 ± 0.086	0.794 ± 0.058	0.802 ± 0.029	0.659 ± 0.013	29.4 ± 1.8%
TVAE	0.904 ± 0.019	0.701 ± 0.036	0.756 ± 0.017	**0.981 ± 0.002**	**0.5 ± 0.2%**

Table 2 External Verification of HIF-Flagged Records against Real-Data Arithmetic Identities ($N = 10$ seeds, mean). Flagged records ($H < 0.5$) are independently confirmed as identity-violating at $\geq 98.9\%$ in every configuration, confirming that the HIF penalty targets genuinely invalid structure. Where both valid and invalid cohorts coexist (Gaussian Copula), unflagged records are markedly less likely to violate (**99.6**% and **72.0**% confirmed vs. 100% among flagged) and the HIF penalty tracks identity severity (Spearman **0.54** and **0.47**).

Dataset	Generator	Flagged	Base Viol.	Conf. @Flagged	Conf. @Unflagged	Spearman ρ
Supermarket Sales						
	Gaussian Copula	33.1%	99.7%	100.0%	**99.6%**	**0.535 ± 0.006**
	Vine Copula	99.7%	100.0%	100.0%	100.0%	0.087 ± 0.010
	CTGAN	99.8%	100.0%	100.0%	100.0%	-0.006 ± 0.027
	TVAE	90.2%	100.0%	100.0%	100.0%	0.459 ± 0.006
Online Purchases						
	Gaussian Copula	54.8%	86.7%	98.9%	**72.0%**	**0.465 ± 0.008**
	Vine Copula	97.2%	100.0%	100.0%	100.0%	0.120 ± 0.018
	CTGAN	96.5%	99.8%	99.9%	98.4%	0.058 ± 0.020
	TVAE	100.0%	100.0%	100.0%	100.0%	0.179 ± 0.011

0.937 vs. 0.940), whereas full HIF filtering raises CTGAN F1 to 0.532 at 65.7% retention. The utility recovery is therefore driven by the LSE/NIC conditional-dependency layers rather than by hard-rule pruning alone. On Online Purchases with the Gaussian Copula (40% violation rate), filtering retains 59.7% of records and F1 moves within noise (0.977 → 0.968), tempering the hypothesis that dependency violations always act as predictive noise: whether removing them improves utility depends on the strength of the violation-to-label association.

The utility-recovery conclusion is robust to the operating threshold, not merely to the continuous score (Table 7). Sweeping the frontier $H \in \{0.3, 0.5, 0.7\}$ on the Census ACS CTGAN cohort ($N = 10$ seeds) yields F1 of 0.723, 0.531, and 0.451 at 30.9%, 65.7%, and 92.6% retention, all significantly above the unfiltered 0.427 (paired t-test $p \leq 0.020$ and Wilcoxon $p \leq 0.037$; 95% CIs $[+0.215, +0.375]$, $[+0.049, +0.160]$, $[+0.005, +0.042]$): the recovery effect increases monotonically with filtering strictness and remains significant even at the loose $H = 0.7$ operating point, where only the most egregious records are removed.

Table 3 Downstream utility filtering across all configurations. Retained cohort size (Retention) dictates whether utility can be reliably estimated. $N = 10$ seeds.

Dataset	Generator	Retention (%)	F1 (Full)	F1 (HIF Filtered)	Δ F1
Adult	CTGAN	70.6	0.449 ± 0.014	0.455 ± 0.029	+0.006
	Gaussian Copula	89.4	0.588 ± 0.035	0.581 ± 0.034	-0.007
	TVAE	99.5	0.741 ± 0.016	0.738 ± 0.024	-0.002
	Vine Copula	84.1	0.459 ± 0.023	0.466 ± 0.021	+0.007
Census ACS	CTGAN	65.7	0.427 ± 0.090	**0.532 ± 0.139**	**+0.104**
	Gaussian Copula	99.2	0.940 ± 0.006	0.937 ± 0.006	-0.003
	TVAE	98.5	0.928 ± 0.011	0.928 ± 0.010	-0.001
	Vine Copula	76.8	0.477 ± 0.039	**0.672 ± 0.037**	**+0.195**
Credit	CTGAN	32.4	0.455 ± 0.021	0.463 ± 0.027	+0.008
	Gaussian Copula	69.5	0.485 ± 0.024	**0.465 ± 0.027**	**-0.021**
	TVAE	86.5	0.573 ± 0.023	0.566 ± 0.031	-0.007
	Vine Copula	44.3	0.439 ± 0.005	0.437 ± 0.004	-0.002
Purchases	CTGAN	15.1	0.372 ± 0.084	**0.500 ± 0.163**	**+0.128**
	Gaussian Copula	59.7	0.977 ± 0.010	0.968 ± 0.024	-0.010
	TVAE	1.1	N/A*	N/A*	-
	Vine Copula	13.8	0.478 ± 0.104	**0.877 ± 0.097**	**+0.399**
Supermarket	CTGAN	0.0	N/A*	N/A*	-
	Gaussian Copula	97.4	0.998 ± 0.002	0.998 ± 0.002	+0.000
	TVAE	0.0	N/A*	N/A*	-
	Vine Copula	0.1	N/A*	N/A*	-

* Retained cohort too small (< 2%) for reliable utility estimation. Bold Δ F1 values are statistically significant (paired t-test, $p < 0.05$).

To characterize when integrity filtering recovers utility across domains, we tested the paired F1 difference (HIF-filtered minus full synthetic) across all dataset–generator combinations of the cross-architecture benchmark with $N = 10$ seeds. Significant utility recovery occurs in four configurations: Census ACS (ΔF1 $= +0.104$, 95% CI $[+0.049, +0.160]$, $p = 0.002$ for CTGAN; $+0.195$, CI $[+0.170, +0.220]$, $p < 0.001$ for Vine Copula) and Online Purchases (ΔF1 $= +0.128$, CI $[+0.010, +0.245]$, $p = 0.037$ for CTGAN; $+0.399$, CI $[+0.306, +0.493]$, $p < 0.001$ for Vine Copula). Recovery is selective rather than a deterministic consequence of the violation rate: high-violation cohorts on Credit Default show no significant change despite 67.6% and 55.7% violation rates (CTGAN, ΔF1 $= +0.008$, $p = 0.29$; Vine Copula, -0.002, $p = 0.16$), and

Online Purchases Gaussian Copula shows none at 40.3% (-0.010, $p = 0.28$). Whether removing flagged records improves utility therefore depends on the strength of the association between the violated conditional structure and the downstream target, not on the violation rate alone. Conversely, most remaining configurations exhibit small, non-significant changes (e.g., Census ACS Gaussian Copula, ΔF1 $= -0.003$, $p = 0.13$; Adult TVAE, -0.002, $p = 0.48$; Adult Gaussian Copula, -0.007, $p = 0.22$), confirming that filtering does not penalize well-formed synthetic data. Only one configuration shows a small but significant degradation (Credit Gaussian Copula, ΔF1 $= -0.021$, CI $[-0.036, -0.006]$, $p = 0.013$), indicating that pruning can also discard flagged records that retain predictive signal for the target; this remains the exception rather than the rule. Finally, four transaction configurations (Supermarket Sales CTGAN/TVAE/Vine; Online Purchases TVAE) retain at most $\approx 2\%$ of records, leaving the retained cohort too small to estimate utility reliably, and their F1 deltas are consequently not reported. Together these results indicate that HIF-driven filtering is a targeted remediation: it recovers utility where structural errors degrade the target signal and is otherwise utility-neutral.

Because sixteen dataset–generator configurations are tested at $\alpha = 0.05$, we report a multiple-comparison sensitivity check in addition to the individual paired tests. Under a Bonferroni correction ($\alpha = 0.05/16 \approx 0.003$), the two Vine Copula recoveries (both $p < 0.001$) and the Census ACS CTGAN gain ($p = 0.002$) remain significant, whereas the Online Purchases CTGAN gain ($p = 0.037$) and the Credit Default Gaussian Copula degradation ($p = 0.013$) fall below the corrected threshold and should be read as suggestive rather than conclusive. The aggregate pattern is nevertheless unlikely to arise by chance: five of sixteen tests reach $p < 0.05$ versus roughly one expected under a global null, and the recovered effects concentrate precisely on the cohorts with the strongest dependency violations.

4.3 Comparison with Outlier Detection and Learned-Density Baselines

A natural question is whether standard outlier detection methods—Isolation Forest (IF) and Local Outlier Factor (LOF)—or a learned-density baseline can serve as alternatives to HIF. Table 4 compares detection performance across five error types on Census ACS at 40% corruption ($N = 10$ seeds); Table 5 replicates the identical protocol on Online Purchases. LOF and IF are configured with a `contamination` parameter set to the known corruption rate (0.4), giving them a tuned decision threshold for F1 in this stress-test setup, whereas HIF uses a fixed non-parametric default threshold ($H > 0.5$). As a learned-density baseline we fit a Gaussian Mixture Model (GMM) on the real data with component count selected by BIC and score synthetic records by negative log-likelihood; its F1 threshold is the $(1-0.4)$ quantile of the real-data scores, the density analogue of the oracle `contamination` calibration granted to IF and LOF. For a like-for-like comparison we additionally report HIF† and GMM†, which flag the top 40% of records by their respective scores, matching the calibration granted to the baselines; in practice the baselines flag a comparable fraction of records (IF $\approx$ 30–42% across error families, LOF $\approx$ 79%), so the matched HIF operating point is not laxer than theirs.

Under threshold-independent ROC-AUC on Census ACS, HIF strongly outperforms the geometric baselines on *Conditional Dependency Violations* (ROC-AUC = 0.830, PR-AUC = 0.789), whereas Isolation Forest is near-random on marginal-preserving dependency swaps (ROC-AUC = 0.512) and LOF achieves lower ranking quality (ROC-AUC = 0.719, PR-AUC = 0.654). A learned-density baseline is competitive with HIF on this family (GMM ROC-AUC = 0.820, PR-AUC = 0.797) but trails HIF at the matched operating point (F1 0.710 vs. 0.720; Table 4, Panel B). The F1 comparison is markedly more favorable to HIF once operating points are matched: flagging the top 40% of records by HIF penalty yields the highest F1 in three of five error families—F1 = 0.720 on dependency violations (vs. 0.593 for LOF, 0.413 for IF, and 0.710 for GMM†), F1 = 0.541 on feature dropout (vs. 0.518 for LOF), and F1 = 0.792 on random noise injection (vs. 0.617 for LOF)—despite flagging only 40% of records to LOF's $\approx$ 79%. Conversely, generic outlier detectors retain the edge on row duplication (LOF F1 = 0.535 vs. 0.406) and global covariate shift (IF F1 = 0.398 vs. 0.163, ROC-AUC 0.490 vs. 0.289), indicating that HIF is a specialized diagnostic for row-level joint conditional integrity rather than geometric redundancy or global distribution drift. HIF also maintains high sensitivity on random noise (ROC-AUC = 0.890) because breaking marginals simultaneously disrupts conditional dependencies. The PR-AUC results are consistent with ROC-AUC across the five families (Table 4, Panel A): HIF leads on dependency violations and feature dropout, while LOF retains the edge on row duplication and IF on covariate shift.

The second-dataset replication on Online Purchases (Table 5) confirms the pattern against the geometric detectors—HIF leads IF and LOF on dependency violations (ROC-AUC 0.797 vs. 0.473 and 0.583), feature dropout (0.716 vs. 0.272 and 0.613), and random noise (0.807 vs. 0.359 and 0.544)—but exposes a clear boundary of the learned-density baseline: GMM is markedly stronger on this domain. On dependency violations GMM attains ROC-AUC 0.955 and PR-AUC 0.909 (vs. 0.797 and 0.627 for HIF) and a matched F1 of 0.891 (vs. 0.665), and it also leads on feature dropout (matched F1 0.736 vs. 0.575) and random noise (0.720 vs. 0.680). We attribute this to the dataset's near-deterministic continuous arithmetic identities (item_total = item_subtotal + item_tax; item_subtotal = purchase_price $\times$ quantity): violating these constraints moves a row sharply off a low-dimensional manifold that a Gaussian mixture approximates almost exactly, whereas HIF's categorical-centric sentinel ensemble has little structure to exploit (a single categorical column) and its NIC residual thresholds are conservatively calibrated to the natural noise floor. HIF's advantage is therefore concentrated precisely where learned-density baselines are weakest: mixed categorical–numeric spaces with complex non-arithmetic conditional structure (Census ACS), in which discrete one-hot support makes Gaussian density estimation unreliable.

Why do standard outlier detectors fail on Dependency Gaps? Isolation Forest and LOF evaluate geometric distance or marginal density across the entire continuous feature space. If a synthetic record hallucinates an impossible logic combination (e.g., a 14-year-old with a PhD) where both values are common marginally, the record resides in a dense region of the globally projected feature space; because Isolation Forest splits features independently, it rarely isolates the specific interacting dimensions that violate the conditional rule. A learned-density model faces a related but distinct limitation:

on the categorical-heavy Census ACS it must place Gaussian components over one-hot encoded indicators, which blurs the sharp conditional structure and dilutes the contrast between a valid record and a dependency-violating one—yet it still comes close to HIF on this family. A Dependency Gap is a structural, semantic error, not a geometric density anomaly, which is why HIF's logic-aware sentinels are competitive with or better than both geometric and learned-density baselines on categorical-conditional dependencies. The replication further shows this advantage is domain-dependent: when the underlying constraints reduce to continuous arithmetic identities, a learned-density model captures them more sharply than HIF's conservatively calibrated residuals, and the two approaches should be treated as complementary—GMM for near-deterministic continuous constraints, HIF for complex categorical-conditional structure, where it additionally provides per-dependency record attribution that a scalar density score does not.

Panel A: ROC-AUC and PR-AUC (↑)

Table 4 Held-Out Error Detection: HIF vs. Baselines (Census ACS, 40% corruption). Panel A reports threshold-independent ROC-AUC and PR-AUC across 10 seeds; Panel B reports F1 at fixed operating points. HIF† and GMM† flag the top 40% of records by their respective scores, matching the oracle `contamination` calibration granted to IF, LOF, and the GMM quantile threshold; unmarked HIF uses the fixed default threshold $H > 0.5$. At a matched operating point HIF attains the highest F1 on conditional dependency violations (0.720 vs. 0.593 for LOF and 0.710 for GMM†), feature dropout (0.541 vs. 0.518), and random noise injection (0.792 vs. 0.617), while the learned-density baseline is competitive at the level of ranking quality (PR-AUC 0.797 on dependency violations vs. 0.789 for HIF).

	ROC-AUC				**PR-AUC**			
Error Type	**HIF**	**IF**	**LOF**	**GMM**	**HIF**	**IF**	**LOF**	**GMM**
Random Noise Injection	**0.890**	0.521	0.864	0.871	**0.852**	0.411	0.829	0.838
Row Duplication	0.506	0.501	**0.508**	0.503	0.407	0.403	**0.409**	0.407
Feature Dropout	**0.652**	0.280	0.489	0.520	**0.582**	0.284	0.397	0.446
Covariate Shift	0.289	**0.490**	0.211	0.276	0.355	**0.411**	0.266	0.284
Conditional Dependency Violations	**0.830**	0.512	0.719	0.820	0.789	0.406	0.654	**0.797**

Panel B: F1 Score at Operating Points (↑)

Error Type	**HIF**	**HIF†**	**IF**	**LOF**	**GMM**	**GMM†**
Random Noise Injection	0.369	**0.792**	0.428	0.617	0.640	0.756
Row Duplication	0.018	0.406	0.404	**0.535**	0.506	0.402
Feature Dropout	0.072	**0.541**	0.157	0.518	0.488	0.415
Covariate Shift	0.007	0.163	**0.398**	0.289	0.327	0.221
Conditional Dependency Violations	0.224	**0.720**	0.413	0.593	0.624	0.710

4.4 Differential Diagnostic: Integrity vs. Loyalty

To determine whether HIF provides unique utility over joint statistical metrics, we evaluated its response at low corruption levels ($p \leq 0.1$) using a conditional-swap protocol that preserves marginal distributions (Census ACS, $N = 3$ seeds). At these low rates, the HIF score decays monotonically with p (Spearman $\rho = -0.98$; from 0.940 at

Panel A: ROC-AUC and PR-AUC (↑)

Table 5 Held-Out Error Detection Replication: HIF vs. Baselines (Online Purchases, 40% corruption). Identical protocol and operating-point conventions as Table 4. On this continuously constrained domain the learned-density baseline is markedly stronger than HIF on ranking quality for *Conditional Dependency Violations* (ROC-AUC 0.955 vs. 0.797; matched F1 0.891 vs. 0.665), while HIF retains an edge over the geometric detectors (IF, LOF) on all three dependency-relevant families.

	ROC-AUC				PR-AUC			
Error Type	**HIF**	**IF**	**LOF**	**GMM**	**HIF**	**IF**	**LOF**	**GMM**
Random Noise Injection	0.807	0.359	0.544	**0.842**	0.701	0.318	0.473	**0.770**
Row Duplication	0.504	0.501	0.505	**0.515**	0.409	0.403	0.406	**0.410**
Feature Dropout	0.716	0.272	0.613	**0.817**	0.566	0.284	0.517	**0.767**
Covariate Shift	0.187	**0.321**	0.275	0.176	**0.344**	0.324	0.310	0.258
Conditional Dependency Violations	0.797	0.473	0.583	**0.955**	0.627	0.373	0.477	**0.909**

Panel B: F1 Score at Operating Points (↑)

Error Type	**HIF**	**HIF**†	**IF**	**LOF**	**GMM**	**GMM**†
Random Noise Injection	0.694	0.680	0.404	0.510	0.606	**0.720**
Feature Dropout	0.624	0.575	0.310	0.544	0.591	**0.736**
Covariate Shift	0.011	0.058	**0.336**	0.288	0.294	0.124
Row Duplication	0.403	0.405	0.508	0.515	**0.550**	0.414
Conditional Dependency Violations	0.720	0.665	0.496	0.552	0.611	**0.891**

$p = 0$ to 0.919 at $p = 0.1$), whereas the aggregate marginal metrics (Moment Matching, KS) remain static and the Joint Correlation Distance moves only marginally ($88.8 \rightarrow 87.9$, relative shift $< 1.2\%$). The small absolute JCD change reflects dilution: the injected rows ($\leq 10\%$ of the 2,000-row cohort) contribute negligibly to a cohort-averaged correlation statistic. Because HIF emits a per-record penalty, it localizes exactly which rows carry the corrupted conditional relationships—a capability no aggregate metric provides. This reading is complementary, not adversarial: under the broad, high-rate permutation protocol described earlier (which destroys pairwise correlations), the JCD decays sharply in lockstep with the HIF score. The two tools therefore address related but distinct failure modes: the JCD is a sensitive dataset-level drift detector, while HIF provides record-level attribution of conditional dependency violations.

4.5 Component Ablation Study

To understand the contribution of each HIF component, we performed an ablation study on Census ACS using CTGAN ($N = 5$ seeds; Table 6). Because Census ACS is dominated by complex, non-linear categorical dependencies, the Logical Sentinel Ensemble (LSE) is the most critical component, raising F1 from 0.422 ± 0.044 (no filtering) to 0.719 ± 0.086 while retaining 27.7% of records. In contrast, rule-based filtering flags few records (2.2% filtered out) and leaves utility near baseline, and NIC-only pruning barely changes retention (13.5% filtered out), since Census ACS numeric features carry few categorical-driven conditional failures; combining LSE with NIC yields intermediate behavior (retention 41.9%, F1 0.639 ± 0.089).

The ablation also illustrates the aggregation method's role: Full HIF with multiplicative product aggregation ($F1 = 0.513 \pm 0.075$) outperforms the arithmetic mean ($F1 = 0.432 \pm 0.047$). The multiplicative product acts as a non-compensatory (Logical AND) gate—preventing a high score in one dimension from hiding a severe failure in another—pruning an additional 30% of records relative to arithmetic aggregation (retaining 64.9% vs. 95.2%). The magnitude of this advantage should be read with caution: retention and utility are confounded, and the gap is driven primarily by how aggressively each variant prunes. Consistent with this interpretation, the aggregation choice is immaterial in settings free of structural errors: under Gaussian Copula on Census ACS (filtered-out rate $< 1\%$), all ablation configurations yield statistically indistinguishable F1 (0.934–0.941).

Why Adding Components Lowers the Flag Rate.

It may seem counterintuitive that adding components (NIC, Rules) to the Logical Sentinel Ensemble (LSE) decreases the flag rate and lowers F1 compared to LSE alone. This behavior stems from aggregating continuous probabilities rather than using boolean rejection. When LSE assigns a harshly low score to a synthetic record, LSE-only will flag it as a violation (leading to a high rejection rate and high F1 on the small remaining cohort). However, if the record is structurally sound in continuous features and rules, NIC and Rule modules will assign it scores near 1.0. Averaging these continuous scores pulls the overall HIF score up, often rescuing the borderline record from falling below the $H < 0.5$ threshold. Consequently, Full HIF acts as a balanced filter that retains a larger, more diverse dataset (yielding an F1 of 0.513 with 64.9% retention) rather than an ultra-strict, low-diversity filter (LSE-only F1 0.719 with 27.7% retention). The geometric mean is used specifically because it is harder to "rescue" a low score via multiplication than arithmetic addition, pruning more aggressively while avoiding total cohort collapse.

Table 6 Component Ablation Study on Census ACS (CTGAN, $N = 5$ seeds). The multiplicative product implements a non-compensatory Logical AND gate, pruning more records than the arithmetic mean and reaching the highest F1 on a cohort with structural errors. In cohorts free of structural errors (Gaussian Copula), all configurations are statistically indistinguishable (F1 0.934–0.941).

Ablation	**Retention (%)**	**F1 (mean ± SEM)**	**Filtered Out (%)**
No filtering	100.0%	0.422 ± 0.044	—
LSE-only	27.7%	0.719 ± 0.086	72.3%
NIC-only	86.6%	0.418 ± 0.044	13.5%
Rules-only	97.8%	0.424 ± 0.041	2.2%
LSE + NIC	41.9%	0.639 ± 0.089	58.1%
Full HIF (arith.)	95.2%	0.432 ± 0.047	4.8%
Full HIF (multiplicative)	**64.9%**	**0.513 ± 0.075**	**35.1%**

5 Discussion

The results indicate that the Hybrid Integrity Framework provides a conservative diagnosis in our experimental settings. By employing a multiplicative aggregation model, HIF ensures that a failure in any single semantic dimension results in a low integrity score. This non-compensatory evaluation is important for high-stakes domains, where a single inconsistent record can invalidate an analysis.

5.1 Reconciling Generative Paradigms: Statistical vs. Neural Trade-offs

A central finding of our cross-architecture audit (Table 1) is that the Dependency Gap is not restricted to deep neural black-boxes; it appears as a recurring pathology affecting both statistical and neural generative models in distinct ways. In our runs, statistical copulas preserve integrity well on Census ACS and Supermarket Sales (HIF 0.94–0.96 for the Gaussian Copula) but degrade on Online Purchases and Credit Default (HIF 0.61–0.63), and Vine Copulas—despite the highest marginal fit (KS $\geq$ 0.969)—catastrophically violate transaction arithmetic constraints (HIF 0.02–0.27). Conversely, neural models such as CTGAN capture discrete categorical clusters on demographic data (HIF 0.33–0.66) but struggle with continuous arithmetic identities (HIF 0.02 on Supermarket Sales). This reconciles our core motivation: the Dependency Gap does not stem from neural opacity alone, nor from copula rigidity. It is an inherent artifact of standard generative loss functions, which optimize aggregate distributional fit (likelihood, Wasserstein distance, or marginal discrimination) rather than row-level joint conditional integrity—so post-hoc row-level auditing is necessary regardless of whether the underlying generator is statistical or neural.

5.2 Practical Interpretation, Specificity, and Scalability

Because HIF computes the geometric mean of the component validities across L auditing layers, the overall score $H(\mathbf{x}) = \left(\prod_l \mathcal{C}_l\right)^{1/L}$ requires simultaneous joint satisfaction of all conditional models—an intentional conservative design for high-stakes auditing. This specificity–sensitivity trade-off is deliberate: HIF prioritizes *dependency specificity*, flagging violations of learned conditional relationships rather than marginal distributional jitter. A generic anomaly detector sensitive to marginal outliers (such as LOF) would produce false positives on valid but rare records; HIF's sensitivity is bounded by the confidence floors learned from the real data distribution.

From a deployment standpoint, integrity filtering is a curation step whose net value must be weighed against retention: in the two strongest recovery settings the retained cohort falls to $\approx$ 14% of the synthetic data (Online Purchases Vine 13.8%, CTGAN 15.1%), and utility gains are selective (significant in four of sixteen configurations). Practitioners should therefore treat HIF primarily as a diagnostic to guide generator refinement and reserve exclusionary filtering for applications that tolerate cohort shrinkage, in line with our recommendation in the broader-impact discussion. **Deployment guidance:** at the default $H < 0.5$ frontier, filtering improves utility only when the violation-to-target association is strong (Census ACS CTGAN/Vine;

Online Purchases CTGAN/Vine); otherwise it is utility-neutral or harmful (Credit Gaussian degradation $p = 0.013$). Retention below $\approx 15\%$ should trigger manual review before deployment.

The LSE architecture is highly scalable, with a computational complexity of $O(N \cdot K)$ for auditing N records across K hubs, and scales effectively to tables with thousands of columns. Our empirical sweep (Appendix D.1) shows stable performance even at $K = 1$, meaning the hub count can be minimized to reduce the K-multiplier overhead. In our benchmarks, auditing 10,000 synthetic rows required $\approx 1.5\,\mathrm{s}$ on standard multi-core commodity hardware. However, the memory overhead of training and storing K Random Forest sentinels for ultra-high-dimensional tables (e.g., $D > 10,000$) remains a potential hardware bottleneck; in such regimes, sparse-linear or neuro-inspired models offer a better trade-off between semantic depth and memory usage.

5.3 Limitations and Failure Regimes

Despite strong results, several limitations remain. First, HIF is not designed as a global covariate-shift detector and can underperform methods such as Isolation Forest on broad geometric drift. Second, its advantage over learned-density baselines (e.g., GMM) is domain-dependent: on datasets whose constraints reduce to near-deterministic continuous arithmetic identities, a density model captures violations more sharply than HIF's conservatively calibrated residuals, and practitioners with such domains should prefer it. Third, the quality of hub selection and confidence-floor calibration depends on sample size and class balance; in sparse regimes, rare-but-valid records may overlap with high-penalty regions. Fourth, the fixed frontier ($H < 0.5$) is practical but not universally optimal; deployment settings may require task-specific threshold calibration and cost-aware tuning, and extending calibration to non-exchangeable data streams via conformal methods [40] remains open. Fifth, the current empirical benchmark set, while diverse, does not exhaust domain types (for example, temporal or relational tables), so external validity should be interpreted cautiously.

5.4 Broader Impact

By providing a verifiable integrity signal, HIF strengthens the case for reproducible research in privacy-sensitive domains. We recommend using HIF primarily as a diagnostic tool to guide generative model refinement rather than as a purely exclusionary filter.

6 Conclusion

As generative models grow in complexity, the metrics used to evaluate synthetic tabular data must evolve beyond aggregate marginal alignment to assess row-level joint distributional consistency. We introduced the Hybrid Integrity Framework (HIF), a probabilistic diagnostic for detecting inter-feature dependency violations in synthetic tabular data, combining the Logical Sentinel Ensemble (LSE) for categorical conditional dependencies with Neighbor-Invariant Continuity (NIC) for continuous feature plausibility within the learned categorical context.

Our empirical results on Census ACS and transaction benchmarks show monotonic HIF degradation with increasing targeted dependency corruption in our experiments, while marginal metrics such as Moment Matching and KS remain static under the same corruption; joint correlation metrics also respond to broad corruption, but only HIF attributes it to specific records. Our cross-architecture audit reveals a persistent "Integrity-Fidelity Decoupling": sophisticated models such as Vine Copulas can achieve high distributional fidelity (KS > 0.96) while failing 85–100% of dependency integrity tests on constrained datasets. By implementing non-compensatory probabilistic aggregation across independently learned dependency models, HIF provides a conservative row-level signal of joint distributional consistency—conformity to the conditional feature dependencies of the real data—that aggregate marginal metrics often miss. We further show that HIF outperforms standard geometric outlier detectors (Isolation Forest, LOF) on conditional dependency violations while remaining appropriately insensitive to marginal noise that does not constitute a dependency failure, and that this pattern replicates on a second dataset. A learned-density baseline (GMM) is competitive with HIF on categorical-heavy Census ACS and surpasses it on continuously constrained transaction domains, so HIF's distinctive value is concentrated in mixed categorical–numeric spaces with complex conditional structure, where it additionally provides interpretable per-dependency attribution that a scalar density score does not. Future work will extend this framework to high-frequency financial time-series data [41] and explore kernel-based projections to capture higher-order dependencies [42–44].

Code and Data Availability

The complete open-source implementation, experimental benchmarks, and dataset configurations are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

Declarations

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Data Availability. The experimental benchmarks and dataset configurations, together with the complete open-source implementation, are available at https://github.com/ahmed-fouad-lagha/tabular-polygraph.

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A Formal Results for the HIF Score Under Stated Assumptions

We provide a formal analysis of Hybrid Integrity Framework (HIF) score properties under the assumptions stated below. The results should be interpreted as assumption-scoped statements rather than unconditional guarantees.

Theorem 1 (Intersection Soundness). The HIF score $H(\mathbf{x})$ defines a membership certificate for the intersection of semantic manifolds $\mathcal{M}_{total} = \bigcap_l \mathcal{M}_l$. Under the idealization that each component validity satisfies $\mathcal{C}_l(\mathbf{x}) = 1$ iff $\mathbf{x} \in \mathcal{M}_l$ and $\mathcal{C}_l(\mathbf{x}) \in [0, 1)$ otherwise, $H(\mathbf{x}) = \left(\prod_l \mathcal{C}_l(\mathbf{x})\right)^{1/L}$ attains its maximum value 1 exactly on $\mathcal{M}_{total}$:

$$\nVdash[H(\mathbf{x}) = 1] = \prod_l \nVdash[\mathbf{x} \in \mathcal{M}_l] \tag{10}$$

Proof. *Completeness:* if $\mathbf{x} \in \mathcal{M}_{total}$ then $\mathcal{C}_l(\mathbf{x}) = 1$ for all l, so $H(\mathbf{x}) = 1$. *Soundness (non-compensatory property):* if $\mathbf{x} \notin \mathcal{M}_{total}$, there exists l^* with $\mathcal{C}_{l^*}(\mathbf{x}) < 1$, and since all validities lie in $[0, 1]$, the geometric mean satisfies $H(\mathbf{x}) < 1$. *Strict convergence:* an "Absolute Violation" in any layer ($\mathcal{P}_{l^*} \to 1$) drives $H(\mathbf{x}) \to 0$ regardless of other layers. $H(\mathbf{x})$ thus behaves as a conservative diagnostic for joint satisfaction of latent manifold constraints, matching the implementation in Algorithm 2.

Theorem 2 (Asymptotic Convergence of the HIF Penalty). Let $\hat{P}_N(x_h \mid \mathbf{x}_{\setminus h})$ be the empirical conditional probability estimator (the Sentinel f_h) trained on a ground-truth dataset $\mathcal{D}_{real}$ of size N. Assuming the base estimator is statistically consistent (e.g., consistent non-parametric trees), as $N \to \infty$, $\hat{P}_N$ converges in probability to the true conditional distribution $P^*(x_h \mid \mathbf{x}_{\setminus h})$, and the empirical penalty $\mathcal{P}_h^{(N)}(\mathbf{x})$ converges pointwise in probability to the corresponding population penalty:

$$\lim_{N\to\infty} P\left(\left|\mathcal{P}_h^{(N)}(\mathbf{x}) - \mathcal{P}_h^*(\mathbf{x})\right| > \epsilon\right) = 0 \quad \forall\epsilon > 0,\ \mathbf{x} \in \mathcal{M}_{real} \tag{11}$$

Proof sketch. *(i)* Because the LSE leverages consistent non-parametric estimators, the empirical conditional probabilities converge pointwise in probability to the true Bayesian posterior P^* (provided the feature space is bounded and leaf nodes grow sub-linearly with N); the pointwise statement reflects that sup-norm convergence is not generally guaranteed for forest-based estimators. *(ii)* Defining the confidence floor δ_h as the empirical 5th-percentile of the sentinel's out-of-bag probabilities yields an asymptotic upper bound of approximately $\alpha = 0.05$ on false positives under calibration assumptions. *(iii)* Records penalized by the converged HIF ($\mathcal{P}^* > 0$) are expected to lie in lower-probability conditional regions of the manifold, supporting a structural-violation interpretation. *(iv)* By the Uniform Law of Large Numbers and the Continuous Mapping Theorem, the empirical Feature Dependency Score $S^{(N)}(h)$ converges in probability to the Bayes optimal classification accuracy $S^*(h)$, implying asymptotically stable hub ranking.

Theorem 3 (Type I Error Bound & Cohort-Purity Approximation). Let $\mathbf{x} \sim P_{real}$ be a valid record on the ground-truth manifold $\mathcal{M}_{real}$ (including rare tail outliers). Under quantile calibration ($\delta_h = Q_\alpha$ and continuous tolerance bandwidth $\gamma_y = Q_{1-\beta}$), the false-rejection probability is upper-bounded by the Union Bound:

$$P\left(H(\mathbf{x}) < 0.5 \mid \mathbf{x} \in \mathcal{M}_{real}\right) \leq \sum_{h\in\mathcal{H}} \alpha_h + \sum_{y} \beta_y \tag{12}$$

Furthermore, let a generative model produce a noisy mixture $P_{syn} = (1-\eta)P_{real} + \eta P_{hallucinated}$, where $\eta \in [0,1]$ is the structural hallucination rate. Rejection filtering with threshold $H \geq 0.5$ yields an approximate retained-cohort purity:

$$\text{Purity}(H \geq 0.5) = \frac{(1-\eta)(1-\alpha)}{(1-\eta)(1-\alpha) + \eta\beta_{hallucination}} \tag{13}$$

where $\beta_{hallucination}$ is the leakage probability of hallucinated records past the filter. For high hallucination rates ($\eta \approx 0.66$), the pre-filter purity is $(1-\eta) = 34\%$; with a calibrated leakage rate $\beta_{hallucination} \approx 0.02$ and $\alpha = 0.05$, filtering elevates purity to over 96% in this illustrative model, theoretically accounting for the downstream predictive utility improvements observed.

B End-to-End Auditing Procedure

Algorithm 1 summarizes the complete manifold calibration procedure—hub selection, confidence-floor calibration, NIC regressor training, and rule mining—and Algorithm 2 details the end-to-end auditing procedure for scoring a synthetic dataset with the trained HIF components.

Algorithm 1 HIF Manifold Calibration

Input: Ground-truth data $\mathcal{D}_{real}$, Hub count K_{hub}
Output: Sentinels $\{f_h\}$, Floors $\{\delta_h\}$, NIC Regressors $\{r_y\}$, Thresholds $\{\tau_y\}$

```
H ← Select top K_hub features by Feature Dependency Score S(h)
for each hub h ∈ H do
    Train Sentinel f_h : X_{\h} → [0,1]^{|X_h|} on D_real to predict x_h
    δ_h ← max(Percentile_5({f_h^OOB(x_h | x_{\h}) | x ∈ D_real}), 0.01)
end for
Z ← SVD(Scale(OneHot(D_cat)))                        ▷ Non-Centered Spectral Manifold
for each continuous feature y do
    Train Gradient Boosting regressor r_y : Z → ℝ on D_real
    E_y ← {|y − r_y(z)| | x ∈ D_real}                  ▷ Ground-truth residuals
    τ_y ← max(Q_0.95(E_y), Median(E_y) + 2 · MAD(E_y))  ▷ Robust Threshold
    γ_y ← max(Q_0.98(E_y), 2 · τ_y, 3 · MAD(E_y), 0.1)  ▷ Dynamic Gamma Scaling
end for
R ← MineImplicationRules(D_real)                     ▷ Apriori-style, min_conf=0.95,
    min_supp=0.005
```

C Logical Sentinel Ensemble (LSE) Implementation Details

On the U.S. Census ACS dataset, the Feature Dependency Score $S(h)$ identified the following top-5 hubs: `household_income`, `housing_cost`, `poverty_status`, `education`, and `tenure`. The confidence floor δ_h is the 5th-percentile of the *out-of-bag* probability mass that a sentinel assigns to the correct category in the ground-truth data (safety floor 0.01); out-of-bag predictions yield out-of-sample floors without a held-out split, reflecting generalization rather than memorized fit, and avoid the instability of 1st-percentile thresholds in sparse data [6, 45]. Hub selection is deterministic: sentinel discovery uses a fixed random state with non-shuffled cross-validation folds, so the hub set depends only on the ground-truth data and is verified identical across all $N = 10$ held-out seeds for every benchmark dataset (e.g., `household_income` through `tenure` on Census ACS; `quantity` through `item_tax` on Online Purchases).

D Extended Results and Sensitivity Analysis

D.1 Hyperparameter Sensitivity

We evaluated HIF robustness across its three key hyperparameters on the Census ACS dataset (2,000 records with 5% injected violations; $N = 5$ seeds, mean $\pm$ SEM). The continuous HIF score is the primary metric; the violation rate derives from applying

Algorithm 2 HIF Synthetic Data Audit

Input: Synthetic record $\mathbf{x}$, Trained components from Alg 1

Output: Record Integrity Score $H(\mathbf{x})$

Step 1: Logical Sentinel Audit (Categorical)
for each hub $h \in \mathcal{H}$ **do**
  $\mathcal{P}_h \leftarrow \text{clip}\left(\frac{\delta_h - f_h(\mathbf{x}_h | \mathbf{x}_{\setminus h})}{\delta_h}, 0, 1\right)$
end for
$\mathcal{P}_{cat} \leftarrow 1 - \prod_{h \in \mathcal{H}} (1 - \mathcal{P}_h)$ ▷ Multiplicative violation aggregation
Step 2: Neighbor-Invariant Continuity Audit (Continuous)
$\mathbf{z} \leftarrow \phi(\mathbf{x}_{cat})$ ▷ Apply learned standardized projection
for each continuous feature y **do**
  $\rho_y \leftarrow |y - r_y(\mathbf{z})|$
  $\mathcal{P}_y \leftarrow \text{clip}\left(\frac{\rho_y - \tau_y}{\gamma_y}, 0, 1\right)$
end for
$\mathcal{P}_{cont} \leftarrow \max_y \mathcal{P}_y$
Step 3: Integrity Aggregation
$\mathcal{P}_{rule} \leftarrow 1$ if $\mathbf{x}$ violates any $r \in \mathcal{R}$ else 0
$H(\mathbf{x}) \leftarrow \left(\prod_l (1 - \mathcal{P}_l)\right)^{1/L} \in [0, 1]$ ▷ Geometric mean of validities

the violation threshold to row-level penalties. The HIF score is stable across hub counts (varying by $< 1\%$ from 0.979 ± 0.0003 at $K = 1$ to 0.987 ± 0.0007 at $K \geq 10$), since geometric-mean aggregation prevents added sentinels from diluting the score; the hub set saturates at $K = 10$ because Census ACS contains only 10 features. Scores are likewise stable for confidence-floor percentiles 1–5 (0.9848 ± 0.0002 at percentile 1 vs. 0.9841 ± 0.0002 at percentile 5; minor 0.3% drop to 0.9815 ± 0.0002 at the 10th), making the 5th-percentile default a good sensitivity–robustness balance. Finally, the continuous score is invariant to the binary threshold (0.9841 ± 0.0002 for thresholds 0.3, 0.5, 0.7), while the violation rate scales monotonically (0.81% → 0.37% → 0.21%), letting practitioners tune the operating point without retraining. The same threshold sweep extends to the downstream-utility protocol (Section 4.2): the integrity-filtered F1 gain on Census ACS CTGAN is significant at every operating point and increases monotonically with strictness (Table 7).

Table 7 Downstream-Utility Threshold Sensitivity (Census ACS, CTGAN, $N = 10$ seeds, mean). Integrity-filtered TSTR F1 and retention at each frontier threshold H; the unfiltered cohort achieves F1 = 0.427. The recovery effect is significant at every operating point and grows with filtering strictness.

Threshold H	Retained	F1	ΔF1	Paired t p	Wilcoxon p	95% CI for ΔF1
0.3 (strict)	30.9%	0.723	+0.295	< 0.001	0.002	[+0.215, +0.375]
0.5 (default)	65.7%	0.531	+0.104	0.002	0.002	[+0.049, +0.160]
0.7 (loose)	92.6%	0.451	+0.023	0.020	0.037	[+0.005, +0.042]

D.2 Sample-Complexity Floor at HIF's Operating Configuration

These bounds are computable at HIF's operating configuration. The identity/uniformity-testing bound $m \gtrsim \sqrt{S}/\varepsilon^2$ [20, 22] inverted at the per-hub, per-category sample sizes HIF actually conditions on yields per-cell detectability floors $\varepsilon_{\text{cell}} = (S_{\text{cell}}/n^2_{\text{cell}})^{1/4}$, where n_{cell} is the number of real rows with hub value c and S_{cell} is the distinct joint configurations of the non-hub features in that cell. Table 8 reports these values for each benchmark dataset using the auditor's default configuration (5 hubs, 10 quantile bins for continuous features). The cohort-level analogue $\varepsilon_{\text{joint}} = (S_{\text{joint}}/n^2)^{1/4}$ with S_{joint} distinct full-row configurations and $n = 2000$ is also shown.

Table 8 Sample-Complexity Floor: per-cell detectability $\varepsilon_{\text{cell}}$ inverted from the identity-testing bound at HIF's actual hub-conditioned sample sizes (5 hubs per dataset, 10 quantile bins). The worst cell (smallest n_{cell}) drives the floor; aggregate $\varepsilon_{\text{joint}}$ assumes a global uniformity test on the full joint.

Dataset	Rows	Med. n_{cell}	Min n_{cell}	Med. $\varepsilon_{\text{cell}}$	Worst $\varepsilon_{\text{cell}}$	$\varepsilon_{\text{joint}}$
Census ACS	2000	200	198	0.266	0.267	0.150
Adult	2000	5.5	1	0.654	1.000	0.149
Credit	2000	12	1	0.537	1.000	0.150
Online Purchases	664	66	1	0.325	1.000	0.193
Supermarket Sales	1000	100	100	0.316	0.316	0.178

The worst-case floors ($\varepsilon \approx 1.0$) occur where long-tail hub categories leave cells with $n_{\text{cell}} \leq 12$ (Adult `native_country`, Credit `pay_*`, Online Purchases `quantity`). Census ACS, whose hubs (`household_income`, `housing_cost`, `poverty_status`, `education`, `tenure`) all have balanced category counts ($n_{\text{cell}} \geq 198$), achieves $\varepsilon \approx 0.27$; Supermarket Sales' hubs are likewise balanced ($n_{\text{cell}} = 100$, $\varepsilon \approx 0.32$). The cohort-level analogue $\varepsilon_{\text{joint}} \approx 0.15$–$0.19$ is tighter because the full sample $n = 2000$ is used without splitting, but it delivers no per-record attribution. HIF's row-level design therefore trades per-cell detectability for granularity; the multi-hub geometric aggregation and the confidence-floor safety minimum (0.01) are the design elements that dampen the long-tail cells' influence on the final score.

D.3 Scalability and Baseline Comparisons

We benchmarked HIF against the **SHAP Distance** semantic fidelity metric [9], which serves as a state-of-the-art baseline for explainability-aware evaluation. Although the SHAP Distance effectively identifies semantic drift, it suffers from severe scalability bottlenecks. HIF auditing scales linearly, $O(N \cdot K)$, where K is the number of hubs. In our empirical scalability benchmark, fitting the 5-hub LSE auditor on 2,000 real records took 9.7 s, after which scoring 10,000 synthetic records took 1.50 ± 0.15 s and scoring 100,000 records took 10.6 ± 0.5 s on commodity multi-core hardware (fit amortized once; wall-clock averaged over three repeats). The marginal per-10,000-row cost at the 100,000-row scale ($10.6\,/\,10 \approx 1.1\,\mathrm{s}$ per 10k rows) confirms the linear scaling. In contrast, SHAP-based methods require $O(N \cdot D \cdot 2^D)$ operations or expensive approximations, which precludes exact evaluation at this volume. Thus, HIF provides comparable detection accuracy while maintaining the computational efficiency required to scale to high-cardinality datasets with millions of rows.